Weekly Report

Sep 25 – Oct 09

Outline

(A) SUMO implementation

(B) Revisiting suggested changes in the potential

(C) Adaptive control

SUMO Implementation

Constraints

Quantity	**Bound**
Longitudinal acceleration	$-9\ \text{m/s}^2\ \leq a_x \leq\ 3.2\ \text{m/s}^2$
Lateral acceleration	$-3\ \text{m/s}^2\ \leq a_y \leq\ 3\ \text{m/s}^2$
Lateral speed	$-3\ \text{m/s}\ \leq v_y \leq\ 3\ \text{m/s}$
Heading	$\lvert v_y \rvert \leq v_x \tan 15^\circ$
Jerk	None

Margins

Item	Margin
Road edge	Safety margin
Wall continuation	$1/d^2$ then linear
k'	$(1 - l/\lvert\Delta x\rvert) \geq 10^{-4}$
τ floor	$\lvert\tau_x\rvert,\ \lvert\tau_y\rvert \geq 10^{-3}$
Sensing window	$[-20$ m, $+r$ ahead$]$
Gradient step	Central difference, $\varepsilon = 0.05$

Handling of the angled case

Component	Handling
Obstacle–ego collision	Not handled; based on a rectangle drawn using the centre of the front bumper
Road edge–ego collision	Not handled; $W \times L$ based on the body centre
Obstacle potential	Not handled; $W \times L$ based on the body centre

Lateral velocity

Old SUMO code

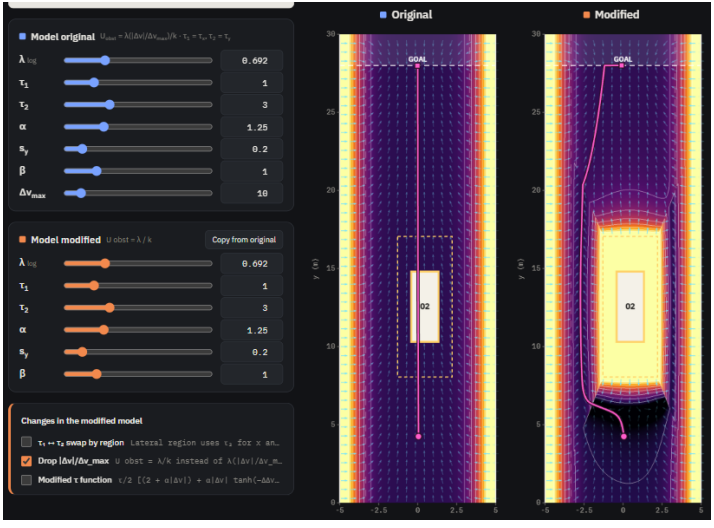
- v_y re-read from SUMO every step
- `getLateralSpeed` $= 0$ for the ego (`moveToXY`)
- $\Delta y = a_y \Delta t^2$

New SUMO code

- v_y kept as controller state
- $v_y = v_{y0} + a_y \Delta t$,

Revisiting Suggested Changes

Analysis through potential fields



CMA-ES Optimisation

CMA-ES: the algorithm

Keeps a **search cloud** $\mathcal{N}(m, \sigma^2 C)$ of parameter sets:

- centre m
- size σ
- shape C

Every generation

1. **Sample**: sample 15 parameter sets from the cloud
2. **Select**: Score each sets and keep the best 7
3. **Move**: centre $\leftarrow$ weighted average of the 7
4. **Update**: σ and C

Differential Evolution (DE)

Keeps a population of parameter sets $x_1, \ldots, x_N$ spread over the search box

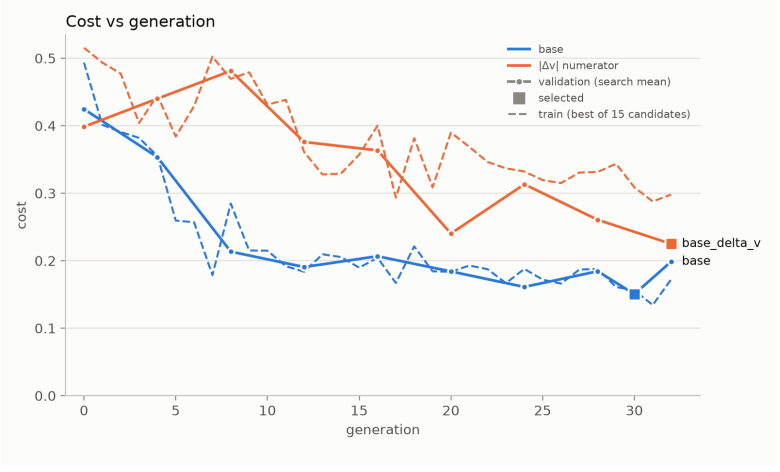
Every generation, for each member x_i

1. **Mutate**: start at the best member, add scaled differences of random members

$$v = x_i + F\left[(x_{r1} - x_{r2}) + (x_{r3} - x_{r4})\right]$$

2. **Crossover**: trial u takes a block of coordinates from v, the rest from x_i
3. **Select**: score u ; u replaces x_i only if it is better

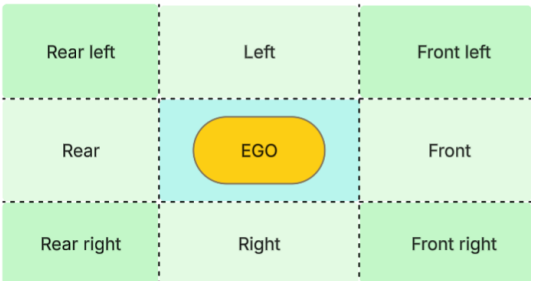
Cost vs generation: base vs $|\Delta v|$ numerator



$$\text{Cost} = 1.1 - (\text{success}) - 0.1(\text{progress})$$

Adaptive Control

Capturing of state



8 features

i) TTC of vehicle ahead
ii) TTC of vehicle front left/right
iii) TTC of vehicles rear (all), sides
iv) Gap front
v) Gap side
vi) Road clearance
vii) Speed
viii) Density

$$f_t = [f_1, f_2, \ldots, f_8], \qquad f_i \in [0, 1]$$

Feature definitions

Feature	Definition
i) TTC ahead	$1 - e^{-1/\mathrm{TTC}}$
ii) TTC front side	$1 - e^{-1/\mathrm{TTC}}$
iii) TTC rear, sides	$1 - e^{-1/\mathrm{TTC}}$
iv) Gap front	$e^{-g_x/20\,\mathrm{m}}$
v) Gap front side	$e^{-g_x/20\,\mathrm{m}}$
vi) Road clearance	$e^{-r/1\,\mathrm{m}}$
vii) Speed	$v_x/34$
viii) Density	$1 - e^{-n/10}$

Adaptive policy

Policy: $W \in \mathbb{R}^{4\times 8}$, $b \in \mathbb{R}^4$ (36 params)

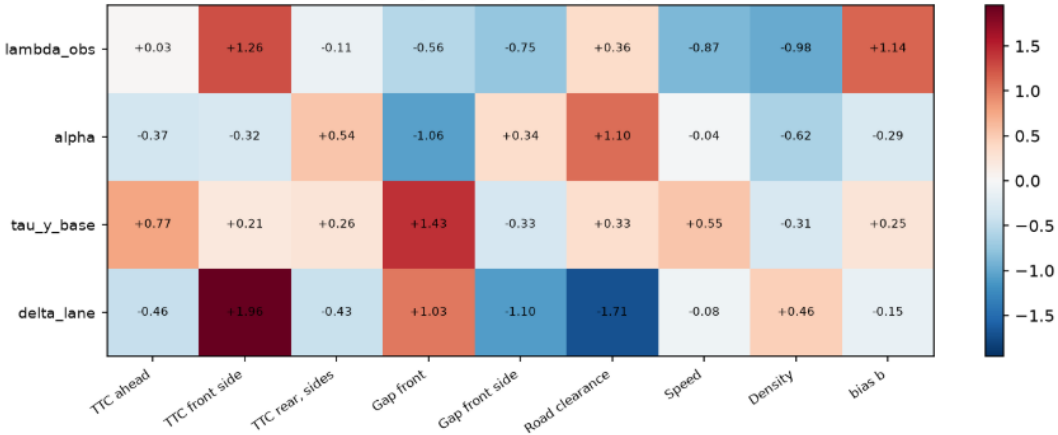
$$z_t^* = \mathrm{clip}(W f_t + b,\ -\ln 10,\ +\ln 10)$$

$$z_t = z_{t-1} + \frac{\Delta t}{T}\,(z_t^* - z_{t-1})$$

$$p_t = p_0\, e^{z_t}$$

$$p = (\lambda_{obs},\ \alpha,\ \tau_y,\ \delta_{lane})$$

Learned policy weights



Training: base vs adaptive

